V6

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Why do we predict the Machine Failure, but NOT the human conditions that cause it?

PREDICT

Because we need signals that we can detect today that help predict a failure tomorrow

Man

Fatigue	Devices that measure the probability of human error. Eye blink rate, slower reaction time, reduced motions.
Weekly seminars	**Mental Health Seminars: Encouraging Employees to Open Up** **Studies indicate that roughly 34% to 55% of employees do not speak up about issues or ideas at work,**

Material problems

We need engineers to ask why did the system allow the mistake

using acoustic sensors to hear tiny crack activity that could jeopardize the production

Improper Methods

Would be teams being reactive rather than being proatcive

PREVENT

Human

Confidence

Material problems

Maintenance

Human-Robot sinergy

improper Methods

No need to ask for WHO

FIND OUT THE WHY

Cybersecurity

Look for a patern = **DATA**

AI As a complement

Implemment with HUMAN expertise

FAST RECOVERY

Human

Empower employees on self diagnosing the problem
Learning through failure

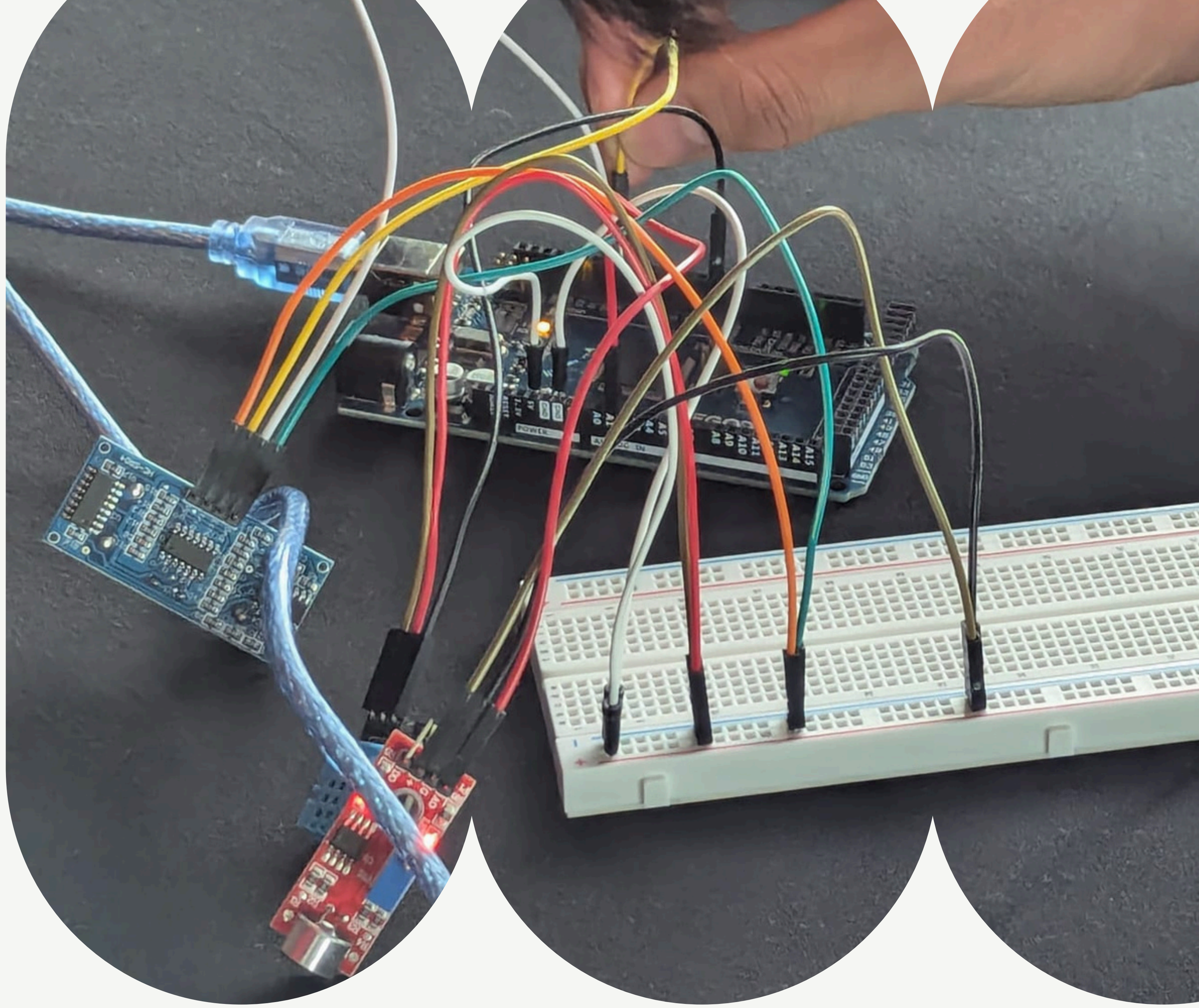
Material problems

AI intelligence to guide the employees to trouble shoot the problem

improper Methods

Hardware Issues & Environmental issues and power outages
-Work Collaboratively

TECHNICAL INNOVATION



PREDICTION **PREVENTION** **RECOVERY**

Prediction & Prevention

OVERALL CONDITION

FAIR

VIBRATION ANOMALY — MIDDLE ARM

Abnormal vibration detected at middle arm — *Inspect middle arm for loose components or bearing wear*

CRITICAL	VIBRATION — MIDDLE ARM	Critical vibration at MIDDLE ARM (delta: 1023) — stop machine immediately	15:10:48
CRITICAL	VIBRATION — TOOL END	Critical vibration at TOOL END (delta: 795) — stop machine immediately	15:10:48
NOTICE	TEMPERATURE — TOOL END	Temp rising at TOOL END: 37.6°C (+11.5°C above baseline)	15:10:48
WARNING	HUMIDITY — TOOL END	High humidity at TOOL END: 26.6% (+19.4% above baseline)	15:10:48

CONDITION REPORT

TEMPERATURE

TEMP — BASE	EXCELLENT
TEMP — JOINT	EXCELLENT
TEMP — TOOL	FAIR

HUMIDITY

HUMID — BASE	EXCELLENT
HUMID — JOINT	EXCELLENT
HUMID — TOOL	FAIR

VIBRATION

VIB — BASE	EXCELLENT
VIB — MID	CRITICAL

LIVE READINGS

TEMPERATURE

BASE	JOINT	TOOL
23.8°C	40.7°C	37.6°C
+0.3 from base	+0.2 from base	+11.5 from base

HUMIDITY

BASE	JOINT	TOOL
45%	22%	26.6%
-1.0 from base	0.0 from base	-19.4 from base

VIBRATION

BASE	MID	TOOL
0	0	121
0.0 from base	-1023.0 from base	-795.0 from base

CONDITION REPORT

TEMPERATURE

TEMP — BASE		EXCELLENT
TEMP — JOINT		EXCELLENT
TEMP — TOOL		FAIR

HUMIDITY

HUMID — BASE		EXCELLENT
HUMID — JOINT		EXCELLENT
HUMID — TOOL		FAIR

VIBRATION

VIB — BASE		EXCELLENT
VIB — MID		CRITICAL
VIB — TOOL		CRITICAL

OTHER

SOUND		EXCELLENT
DISTANCE		FAIR

LIVE READINGS

TEMPERATURE

BASE
23.8°C
+0.3 from base

JOINT
40.7°C
+0.2 from base

TOOL
37.6°C
+11.5 from base

HUMIDITY

BASE
45%
-1.0 from base

JOINT
22%
0.0 from base

TOOL
26.6%
-19.4 from base

VIBRATION

BASE
0
0.0 from base

MID
0
-1023.0 from base

TOOL
121
-795.0 from base

OTHER

SOUND
46
0.0 from base

DISTANCE
9.3cm
+6.0 from base

TEMPERATURE TREND — BASE SENSOR

🔧 PARTS BREAKDOWN

FANUC R-2000iA/165F

Axes: 6 | Payload: 165 kg | Reach: 2655 mm

● Sensor feed offline

J1 — Waist Rotation ●
Aluminum alloy housing
🌡 50.7°C 0.15g ⚡ 40%

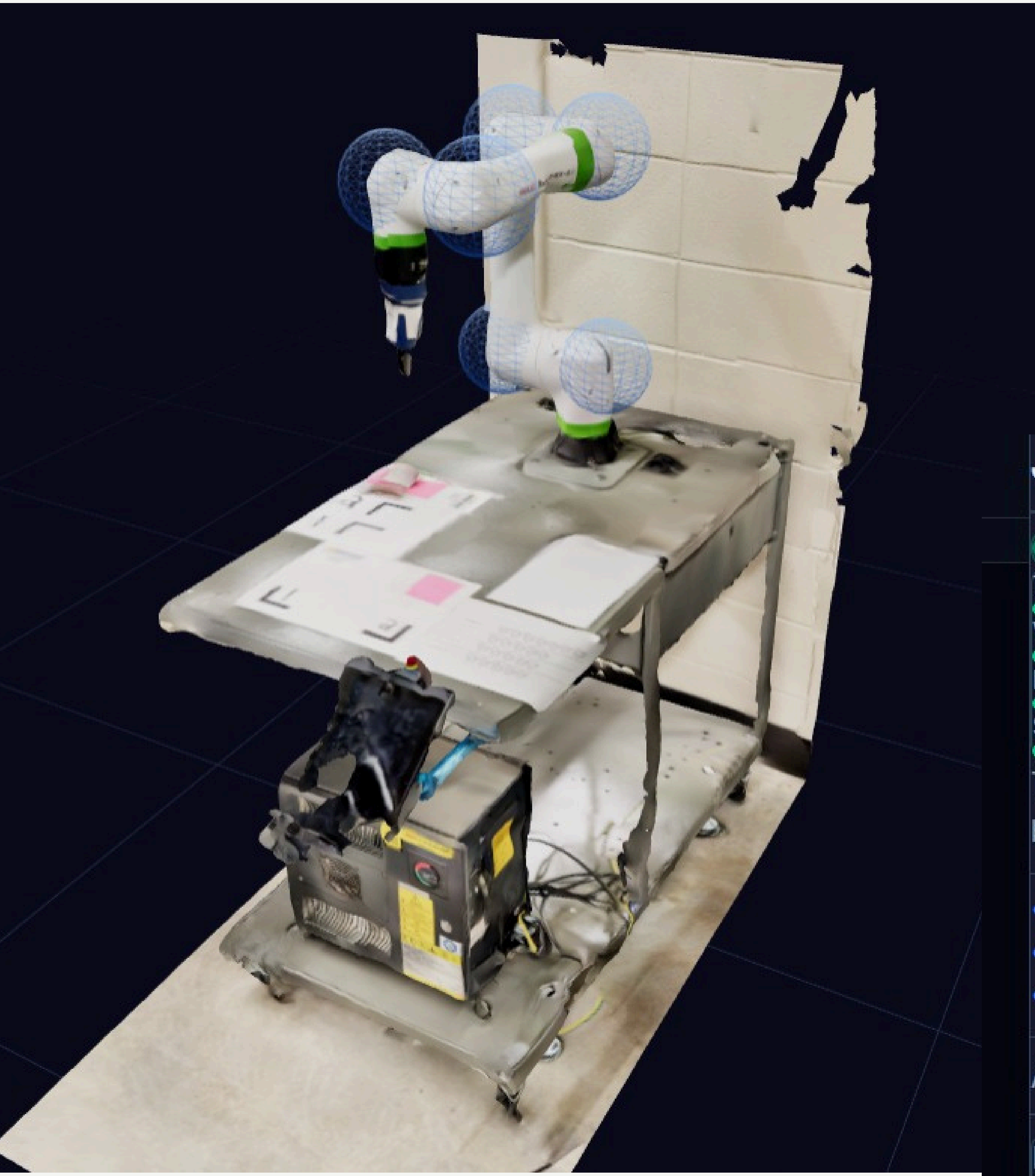
J2 — Shoulder (Lower Arm) ●
High-strength steel
🌡 50.0°C 0.04g ⚡ 68%

J3 — Elbow (Upper Arm) ●
Aluminum alloy
🌡 51.7°C 0.04g ⚡ 37%

J4 — Wrist Roll ●
Aluminum alloy
🌡 36.4°C 0.22g ⚡ 33%

J5 — Wrist Pitch ●
Aluminum alloy
🌡 45.7°C 0.20g ⚡ 69%

J6 — Wrist Yaw (Tool Flange) ●
Steel flange, aluminum housing
🌡 51.3°C 0.10g ⚡ 43%



❗ ERROR LOG

Live & recent exclamation alerts

Live Errors

No live errors.

Recent Errors

No recent errors.

J1 — WAIST ROTATION

LIVE SENSOR DATA

● NORMAL 4:20:43 PM

Temperature	42.9 °C
Vibration	0.2 g
Load	38.6 %
Speed	101.2 °/s

The first axis. Rotates the entire arm left and right around the base. Provides the robot's full rotational workspace.

SPECIFICATIONS

- Rotation range: ±360°
- Max speed: 120°/sec
- Driven by AC servo motor + harmonic drive

MATERIAL

Aluminum alloy housing

MAINTENANCE NOTES

Listen for unusual grinding sounds during rotation. Monitor motor temperature. Grease harmonic drive per schedule.

Fast Recovery

Learnings

- Had to transfer raw data from one device to another using IP transfer.
- Learnt that we wont know which joint will fail, we improvise using 3 putting at -prioritizing major failure points

HUMAN INNOVATION

Keyword

Incentivization structures
- Increase vacation time
- Company Discounts
- Money/ Promotion

Thank you.

Questions?